

## UNIT 5: PRACTICE QUESTION BANK & ANSWER KEY

**Course Pattern:** TE IT - 2024 Pattern (SPPU)  
**Document Type:** Question Bank with Answer Key

**Academic Term:** Semester V (2026)  
**Course Code:** PCC-301-ITT

### Section A: Important Theory Questions

1. **I/O Architecture & Buffering:**

- Compare Programmed I/O, Interrupt-Driven I/O, and Direct Memory Access (DMA). Why is DMA preferred for high-speed data transfer? [SPPU Nov 2023, Marks: 8]
- Explain Single Buffering, Double Buffering, and Circular Buffering in I/O management. [SPPU May 2024, Marks: 6]

2. **Disk Architecture & Scheduling:**

- Explain the components of total Disk Access Time (Seek Time, Rotational Latency, Transfer Time). [SPPU Dec 2022, Marks: 6]
- Compare FCFS, SSTF, SCAN, C-SCAN, and LOOK disk scheduling algorithms. Explain how SSTF can lead to starvation. [SPPU Nov 2023, Marks: 8]

3. **File Allocation & Unix Inode:**

- Compare Contiguous, Linked (FAT), and Indexed file allocation methods with respect to random access speed and fragmentation. [SPPU May 2023, Marks: 8]
- Draw and explain the structure of a Unix Inode. Show how direct, single indirect, double indirect, and triple indirect pointers enable large file support. [SPPU May 2024, Marks: 8]

### Section B: Numerical Practice Problems

#### Practice Problem 1: Disk Head Movement Calculations

A disk queue contains requests for cylinders: 98, 183, 37, 122, 14, 124, 65, 67. The head is currently at cylinder **53**, moving toward **higher cylinders** (0 to 199).

Compute total head movement (in cylinders) using:

- (a) FCFS Disk Scheduling [Marks: 3]
- (b) SSTF Disk Scheduling [Marks: 3]
- (c) SCAN Disk Scheduling [Marks: 3]
- (d) LOOK Disk Scheduling [Marks: 3]

### Mandatory Final Answer Key

Below are the verified solutions for self-verification:

• **Practice Problem 1 — Disk Head Movements:**

- (a) **FCFS:** 53to98to183to37to122to14to124to65to67
  - **Total Movement:**  $45 + 85 + 146 + 85 + 108 + 110 + 59 + 2 = \text{bold}\{640\}$  cylinders

- **(b) SSTF:** 53to65to67to37to14to98to122to124to183
    - **Total Movement:**  $12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 = \text{bold}\{236 \text{ cylinders}\}$
  - **(c) SCAN (Elevator to boundary 199):** 53to65to67to98to122to124to183to199to37to14
    - **Total Movement:**  $(199 - 53) + (199 - 14) = 146 + 185 = \text{bold}\{331 \text{ cylinders}\}$
  - **(d) LOOK (To max request 183):** 53to65to67to98to122to124to183to37to14
    - **Total Movement:**  $(183 - 53) + (183 - 14) = 130 + 169 = \text{bold}\{299 \text{ cylinders}\}$
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*End of Operating Systems Unit 5 Practice Question Bank & Answer Key — SPPU TE IT 2024 Pattern*